Qorium - Technical Case Study - Cost-Aware Cell Culture Media Optimisation

Technical memo · Daan Knoors · October 2026

1. Background

A cell-culture team wants to improve culture performance while reducing media cost. Experiments are expensive, noisy and there is limited historical data. The objective is to recommend the next batch of media formulations to test.

1.1. Data

For this experiment we compared the data provided by Cosenza et al. (2022) with the two datasets from Narayanan et al. (2025). Given the time constraints, we chose to focus on the former, since it better aligns with the objective and conditions of the case study: it measures cell count, includes cost data and both cheap and expensive assays, focuses on mammalian cells, and has 14 continuous components, which seems more realistic for an industrial setting. However, it lacks categorical variables, which would likely exist in a real-world setting and require different techniques.

1.2. Objective

Qorium has already proven that their technology works and is well on its way to transition from an academic setting to an industrial one. The main challenge remains scalability. Hence, the most practical objective focuses on improving cell growth, while reducing media cost.

Objective: maximise cell count subject to a cost ceiling

For the cost ceiling, we set this equal to the commercial baseline model, consisting of DMEM + 10% FBS. The idea is more cell growth for no more money. At a later stage, we also evaluate how much gain in cell count we can achieve if we relax this cost constraint slightly.

1.3. Assumptions & Constraints

Assumptions:

- We focus only on high-fidelity assay results, as the low-fidelity assays can be inaccurate. In future work we may consider integrating these into the model.
- Two-passage results are assumed to predict longer cultures, thus the expensive assays are a good predictor of long term growth.
- Absolute cell numbers are not needed, we only rank recipes and report gains in standard deviations as the data is already pre-processed and standardized.
- Costs can be computed using the paper's linear, relative cost.

Constraints:

- Each component stays between 0 and its maximum concentration listed in the paper.
- Media cost stays at or below the commercial baseline (see 1.2).

1.4. Disclaimer

I do not have a background in cell-culture R&D. Any assumptions made in this document are to be verified with the R&D team. Conclusions may alter significantly based on budget constraints, previous experiment results, and other factors.

Given the time constraints and the complexity of the task, I want to be transparent that I used an AI assistant for this assignment, including for reading the reference papers and for writing most of the code under my direction. I worked through the problem step by step, reviewing and questioning each step until I could explain it, and made the key decisions myself, including the choice of dataset and objective. I checked the results against the paper's published figures, and I can explain and defend every choice in this memo.

2. Methodology

The full methodology is in the notebooks and Python modules of the code repository; we summarise the main points here.

2.1. Pre-processing

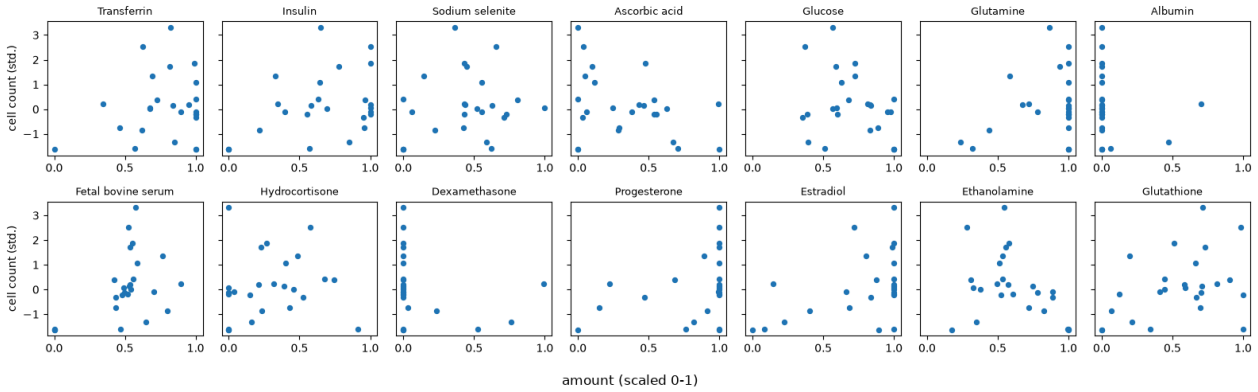
When we look at the data we see it's already pre-processed and standardized. Column names and values were inferred from the tables included in the paper. We verified that the values of the best performing recipe correspond to the reported figures.

For simplicity and accuracy, we only kept the expensive assay results, as these correlate best with our objective of maximising cell count. This leaves us with 25 recipes, each with 14 components. The cheaper assays (AlamarBlue, LIVE stain) correlate only moderately and flatten out for good recipes; we leave them for future work (see 4.3).

2.2. Exploratory findings

We first checked how noisy the measurements are: the standard deviation between replicate wells (0.21) is small compared with the spread between recipes (1.31). This means that the differences between recipes are real and not measurement noise.

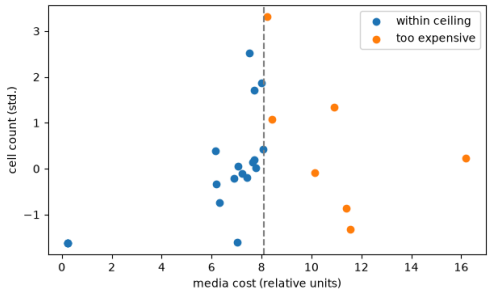
Furthermore, we examined how each component affects the cell count. Below you can find a scatter plot of each recipe's cell count vs. the component concentrations.



While it's hard to generalise with only 25 data points, we can observe some patterns:

- No Fetal Bovine Serum (FBS) results in poor cell growth and high values may actually harm it as well. The best recipes sit around 0.55 of the maximum FBS concentration (about 10%) similar to the commercial baseline.
- Best recipes have no albumin and no dexamethasone, while glutamine and progesterone are close to the maximum for the best recipes.
- No single component explains growth on its own, so a model must use all of them together.

Subsequently, we look at our cost ceiling which we set equal to the commercial baseline, see the figure below.



Only 18 of the 25 recipes are within the ceiling. The best of these scores 2.52. This excludes the best recipe found by the authors with a score of 3.31 costing only 1.6% more than the baseline. We examine the effect of raising or lowering the ceiling in 3.2. Note that a higher cost doesn't necessarily result in a higher cell count.

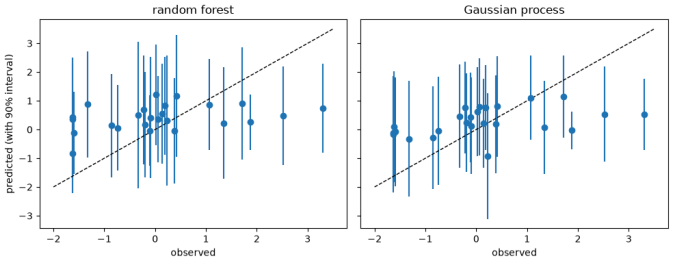
2.3. Modeling

We now train a model that predicts the cell count of a recipe. To balance exploration and exploitation, the model must also give a measure of uncertainty. We compared three models that do so: a simple average (baseline), a random forest and a Gaussian process (GP), which predicts a new recipe from similar measured recipes.

We evaluated the models with leave-one-out cross-validation, where we train on 24 recipes, predict the 25th, repeat for all 25. We would have preferred to dedicate a larger portion to the test data, but given the size and dimensionality of the dataset, we want the model to pick up as many of data's underlying patterns as possible.

Model	Error (RMSE)	Ranking (Spearman)	90 % interval coverage
Average (baseline)	1.33	–	0.92
Random forest	1.22	0.36	0.76 (over-confident)
Gaussian process (GP)	**1.16**	**0.52**	**0.88** (honest)

The Gaussian process wins on every measure: lowest error, best ranking and honest uncertainty, which the selection step depends on. Below, the predictions with their 90% interval vs the observed results.

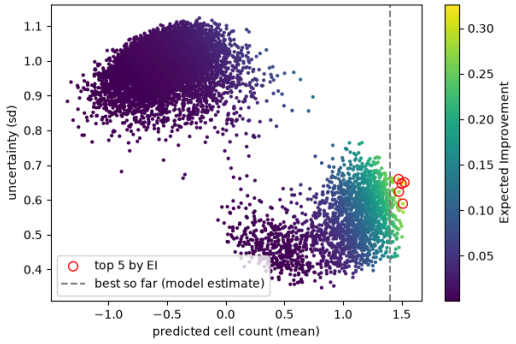


Notice none of the models predict well. There is just not enough data to train them. Both of them underpredict the best recipes. That is why we should not pick the highest prediction, but also use the uncertainty.

2.4. Experiment selection

Similar to the authors, we choose experiments with Bayesian optimisation:

1. **Candidates**: we generate 20.000 random recipes (using Latin Hypercube sampling) as well as 5000 small variations of the 5 best recipes. Subsequently, we only keep those within the cost ceiling. With 14 dimensions, random sampling alone would never reach enough candidates close to the known good region.
2. **Scoring**: we use Expected Improvement (EI), which is the average gain over the best so far and rewards high prediction *and* high uncertainty, thus inherently balancing exploitation and exploration.
3. **Bar**: we set the bar equal to the model's own estimate of the best tested recipe (1.40), not the actual value (2.52). This value would be too optimistic according to the model and could be explained by measurement noise.
4. **Batch**: we select 3 exploit (highest EI) + 1 explore (most uncertain recipe still predicted at least average) + 1 benchmark. Picks are made one at a time, we then pretend this pick was measured already with its predicted value, so the next pick goes elsewhere. We chose 3 + 1, because with only a few experiments, most of them should have a high chance to pay off. Moreover, EI already contains some exploration. The one explore slot remains for the model to learn unexplored regions.
5. **Uncertainty**: besides this explore/exploit balance, the GP's 90% intervals proved honest in leave-one-out (88% of measurements fall within them), and the benchmark in every batch measures drift between batches and flags deviating ones.



3. Results

3.1. Recommendation

Here is the next batch of experiments provided by the model. See the notebook for full recipes in lab units.

#	Role	FBS % v/v	Albumin	Glucose (mg/ml)	Glutamine (mg/ml)	Predicted y [90 % interval]	Cost vs today
1	Exploit	8.6	0	6.0	1.52	1.52 [0.45, 2.59]	−13 %
2	Exploit	7.7	0	7.0	1.52	1.47 [0.39, 2.56]	−21 %
3	Exploit	9.8	0	7.1	1.35	1.43 [0.39, 2.46]	−1 %
4	Explore	1.9	1.13	6.2	1.48	0.01 [−1.81, 1.82]	−30 %
5	Benchmark	10.0	0	4.5	0.43	(reference)	0 %

The **exploit** recipes refine the known region with high predicted cell count: high insulin and progesterone, no albumin or dexamethasone. They vary the FBS, transferrin, and glucose.
The **explore** recipe tests a strategic question: can albumin partly replace FBS. It is uncertain, but with success it would open a much cheaper direction.

3.2. Cost ceiling sensitivity

How much does the cost ceiling influence the recommendation? The table shows the average over the 3 exploit picks.

Ceiling	Predicted y	Pessimistic y	Cost vs today
0.75×	1.37	0.42	0.74× (extrapolation: nothing tested this cheap grew well)
1.0×	**1.47**	**0.41**	**0.88×**
1.05×	1.58	0.60	0.98×
1.25×	1.60	0.58	1.16×

We advise to keep today's benchmark cost as the default. A +5% increase is a reasonable option, as it admits the best recipe found so far with a minimal cost increase. At 25% increase the advantage is minimal.

4. Discussion

4.1. Risks and limitations

- **Tiny data**: data is too limited to make robust predictions and error is high.
- **Biased history**: it was sampled near areas that were already proven to be good.
- **No absolute units**: outputs are standardised so there are no absolute cell counts.
- **Narrow scope**: one cell line, small plates, and a linear cost model. Not reflective of scaling up to industrial settings
- **Batch noise**: noise between batches is not measured and could not be accounted for.

4.2. Validation plan

1. Run the recommended batch with 3 replicate wells per recipe, including the benchmark
2. Judge: If at least one exploit recipe beats the benchmark at equal or lower cost we consider it a success. Also check that the results fall inside the predicted 90% intervals: is the model honest enough on new data?
3. Add the results, refit and recommend again. Stop when gains flatten.
4. Confirm the winner over multiple passages before adopting it.

4.3. Extensions

How our strategy would change in different scenarios:

Scenario	Change
Only 10 -20 prior experiments	Less than 10, skip the model and run a space-filling design (Latin Hypercube). After collecting more data points, use the GP with simple settings, give exploration priority and lean on domain knowledge
Low-fidelity assays are cheap but noisy	These correlate moderately and flatten out for good recipes. Use them to discard poor recipes and spend expensive assays only on promising candidates. Explore a multi-fidelity model that combines both.
High-fidelity assays are expensive but reliable	Focus on exploitation of good candidates. Smaller batches, more rounds.
Component cost data incomplete	Keep the ceiling relative to today's baseline. This is robust against price changes. Use price ranges for unknown items and check whether the recommendation changes. Try to at least get real quotes for the most expensive items.
Categorical or constrained components	Constraints become candidate filters, similar to the cost ceiling. Categories need an encoding or a kernel that compares categories (Narayanan, 2025).
Experiments must run in parallel batches	The model recommends multiple recipes that can be tested in parallel. Size the batch to lab capacity. Larger batches can afford more exploration slots.

5. References

- Cosenza, Zachary, Raul Astudillo, Peter I. Frazier, Keith Baar, and David E. Block. 'Multi-information Source Bayesian Optimization of Culture Media for Cellular Agriculture'. *Biotechnology and Bioengineering* 119, no. 9 (2022): 2447–58. https://doi.org/10.1002/bit.28132.
- Narayanan, Harini, Joshua Hinckley, Rachel Barry, et al. 'Accelerating Cell Culture Media Development Using Bayesian Optimization-Based Iterative Experimental Design'. *Nature Communications* 16 (2025): 6055. https://doi.org/10.1038/s41467-025-61113-5.